



1.4 The binomial theorem

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

For positive integer n , $(a+b)^n = \sum_{r=0}^n {}^nC_r \cdot a^{(n-r)} \cdot b^r$, where ${}^nC_r = \frac{n!}{r!(n-r)!}$ are binomial coefficients (row n of Pascal's triangle). The general $(r+1)$ th term is ${}^nC_r \cdot a^{(n-r)} \cdot b^r$, useful for finding a specific term without expanding fully.

Worked example: follow the reasoning

Find the coefficient of x^3 in the expansion of $(2+x)^5$.

General term: ${}^5C_r \cdot 2^{(5-r)} \cdot x^r$. Need $r=3$: ${}^5C_3 \cdot 2^2 \cdot x^3 = 10 \times 4 \times x^3 = 40x^3$.

Coefficient = 40.

Quick check · multiple choice

What is the value of 6C_2 ?

- A 12
- B 15
- C 30
- D 20

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider the expansion of $(1+3x)^7$.

[7]

- (a) Write down the general term.
- (b) Find the coefficient of x^4 .
- (c) Find the term independent of x in the expansion of $(2x - 1/x)^6$.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) General term: $7Cr \cdot (3x)^r = 7Cr \cdot 3^r \cdot x^r$.

(b) $r=4$: $7C4 \cdot 3^4 = 35 \times 81 = 2835$.

(c) General term of $(2x-1/x)^6$: $6Cr(2x)^{6-r}(-1/x)^r = 6Cr \cdot 2^{6-r} \cdot (-1)^r \cdot x^{6-2r}$.

Independent of x needs $6-2r=0$, $r=3$: $6C3 \cdot 2^3 \cdot (-1)^3 = 20 \times 8 \times (-1) = -160$.

Quick check answer

B. $6C2 = 6!/(2!4!) = (6 \times 5)/2 = 15$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.